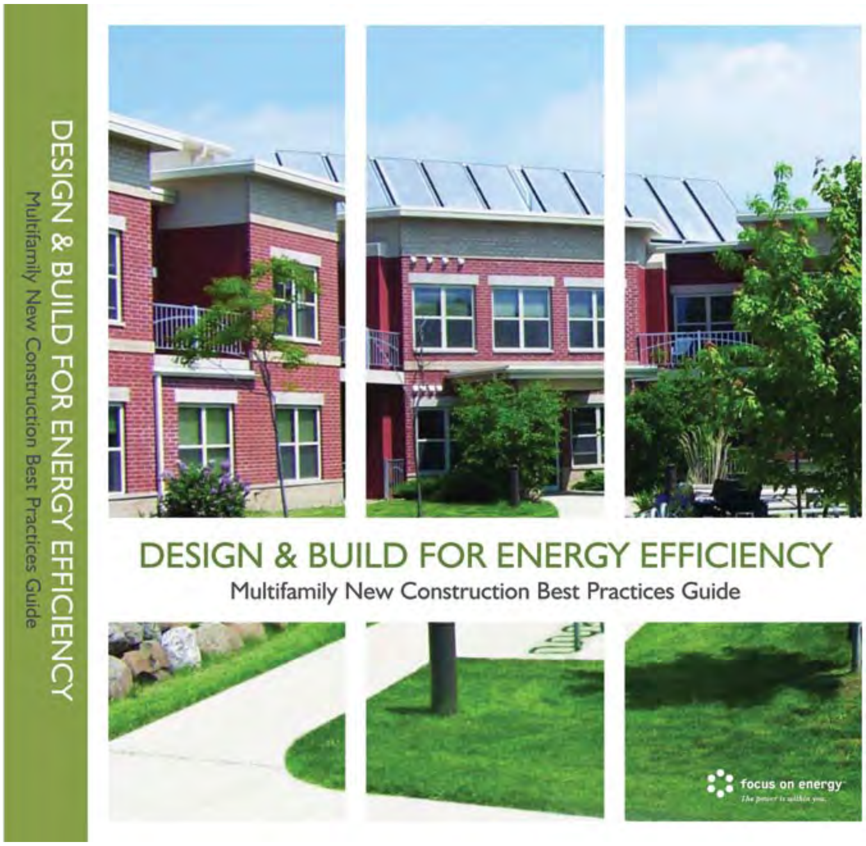
COLLABORATING FOR ENERGY EFFICIENCY

Wisconsin energy efficiency and building organizations partner to create a multifamily manual.

by JOSH ARNOLD

For many households, energy is one of the top monthly expenses. As a response to the high cost of energy over the last few years, several companies dedicated to promoting safe, affordable housing for people throughout Wisconsin have partnered to create a manual that will assist professionals in the multifamily construction industry. The goal of the manual, *Design & Build for Energy Efficiency: Multifamily New Construction Best Practices Guide*, is to inspire the multifamily new-construction industry to achieve energy-efficient, high-performance buildings in a cost-effective manner. It was developed for the express purposes of providing and encouraging voluntary and non-enforced measures for constructing high-performance buildings. *Design & Build* contains technical and business information related to energy efficiency and case studies of projects in Wisconsin that have successfully incorporated Focus on Energy recommendations.



This sleek, easy-to-use guide will benefit everyone involved in the multifamily new construction industry.

Why Use the Guide?

Design & Build was created as a desktop resource for multifamily new-construction professionals. It provides technical information and recommendations to implement cost-effective energy efficiency and renewable energy measures to help professionals build and operate high-performance buildings. Its target audience is all professionals involved with planning, designing, financing, building, or operating apartment and condominium buildings in Wisconsin. Professionals working on projects in climate zones similar to that of Wisconsin may also find the guide a useful resource.

In the guide, we define the high-performance building as one that features

- energy efficiency and renewable energy;
- sustainable site planning;
- safeguarding of water and water efficiency;
- conservation of materials and resources; and
- indoor environmental quality.

High-performance buildings are a good investment for a number of reasons. Initially, high-performance buildings attract investors and capital, and they also help to win neighborhood and local government approval. High-performance buildings can earn higher rents and higher sales value for owners.

They often attract tenants and buyers more quickly, reduce tenant turnover, and reduce liability and risk. The higher indoor environmental quality also ensures that residents will enjoy better health and comfort. In Wisconsin's hot summers and frigid winters, high-performance buildings also cost less to operate and maintain. And of course, all of this also helps to improve the Wisconsin economy.

How Is the Guide Organized?

Design & Build is organized by end user groups, including the owner/developer; architect; engineer (civil, structural, HVAC, electrical, plumbing); and contractor. As a reference tool, it pulls from existing sources of information and expands on key points that may not be well defined in other sources. A sidebar in each section contains specific resources, additional or clarifying information, and recommendations.

Within each section we make an effort to identify specific points of communication and strategies that meet specific goals for efficiency or sustainability through the use of icons. For instance, one icon indicates a strategy to achieve specific goals of sustainability/green, energy efficiency, or indoor air quality. We use a one-to-four scale to indicate level of investment and resulting benefit—one being the least costly method or the method that provides only marginal benefits; four being the most expensive method or the one that provides the greatest benefits.

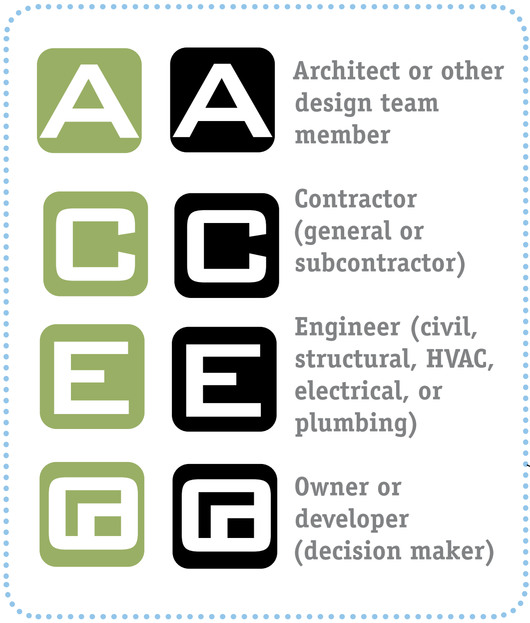
At the end of each section is a case study of a building that participated in the Focus on Energy Multifamily New Construction Program and utilizes various efficiency and sustainable strategies. The case studies provide multifamily professionals with estimated energy efficiency and financial results of incorporating these strategies into the design and construction of a multifamily building. Each section of *Design & Build* is highlighted with sidebars and examples of issues that professionals may encounter during the building process. Building professionals can use this material to make a convincing case to multifamily developers and designers for constructing high-performance buildings. Examples from the guide include

- case studies of multifamily building projects that used design and technical assistance to achieve high-performance goals;
- breakdown of the economics for high-performance multifamily building projects;
- a resources section that lists dozens of organizations and Web sites that are dedicated to promoting energy efficiency and high-performance green buildings both in Wisconsin and throughout the country; and
- design materials for architects including reducing HVAC costs, optimal building orientation, indoor air quality, and sustainable materials and finishes.

Design & Build also includes a Quick-Start Guide to help users identify key decision and communication points that occur during each stage of the design process. It also includes a section on resources. This is not an exhaustive

FOCUS ON ENERGY

The solar panels atop Yahara River View Apartments provides 20%–30% of the average yearly hot water needs.



ENERGY CENTER OF WISCONSIN/FOCUS ON ENERGY

Various icons are used throughout the guide to indicate the importance of information to team members. Other icons indicate strategies to reach specific sustainability, energy-efficiency, and indoor air quality goals as well as coast/benefit scales.

listing; it contains selected resources that we believe to be particularly relevant to multifamily construction.

Design & Build supports design teams by integrating information from various sources. It provides links to ASHRAE's *Advanced Energy Design Guide*, USGBC's LEED V 2.1 and 2.2, and EPA's Energy Star program.

Who Created the Guide?

Focus on Energy is a public-private partnership offering energy information and services to residential, business and industrial customers throughout Wisconsin, with the goals of encouraging energy efficiency and use of renewable energy, enhancing the environment, and ensuring the future supply of energy for Wisconsin.

To create *Design & Build*, Focus on Energy hired the Energy Center of Wisconsin and the New Buildings Institute, the developers of the Advanced Buildings Benchmark. The Advanced Buildings Benchmark provides technical resources, trainings, and information to improve the way buildings are designed, built, and used. Using whole-building patterns, design process tools, and educational resources, the Advanced Buildings Benchmark provides designers with ways to incorporate integrated design strategies into the new construction process to reduce energy use and to improve indoor environmental quality. Advanced Buildings is compatible with both LEED and Energy Star.

Advanced Buildings was developed by the Energy Center of Wisconsin and the New Buildings Institute through the support of their sponsors. The Energy Center of Wisconsin is a private nonprofit organization located in Madison,

FOCUS ON ENERGY

Radiant flooring is combined with solar domestic hot water at the Pheasant Ridge Apartments in Madison, Wisconsin.

FOCUS ON ENERGY

This network of high efficiency sealed combustion modulated boilers results in 75% less heating intensity than a typical apartment in Madison.

Wisconsin. It is dedicated to improving energy sustainability by supporting energy efficiency, renewable energy, and environmental protection. The New Buildings Institute is a nonprofit organization located in White Salmon, Washington. It works with national, regional, state, and utility groups to promote improved energy performance in commercial new construction.

How the Program Works

The Focus on Energy multifamily program provides multifamily professionals with cost-effective energy efficiency ideas that they can incorporate into their new- construction projects at little or no additional cost or effort. We provide three kinds of resources:

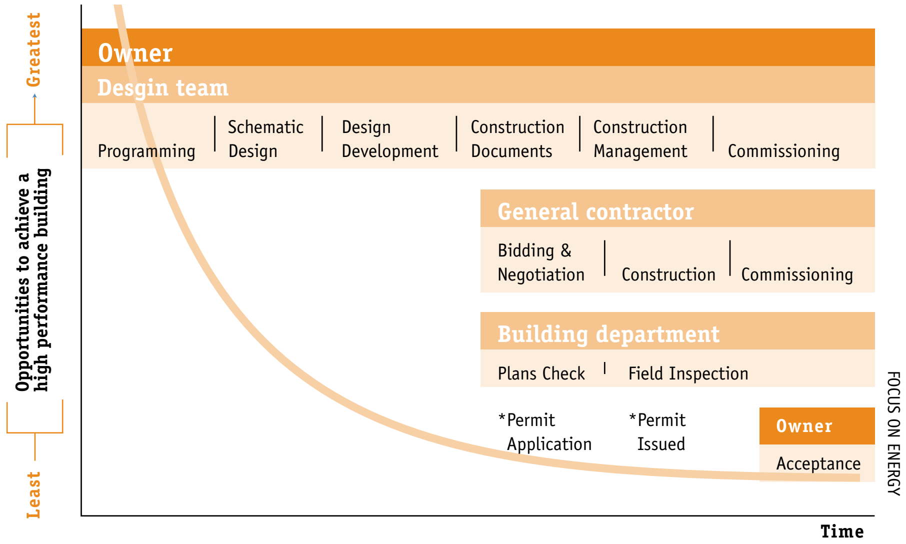
Cash incentives for energy efficiency. A project may qualify for cash incentives based on the number of energy efficiency measures incorporated into the final project. Incentives are based on the amount of energy saved, and are paid after verification that the energy efficiency measures are installed at the final project. Typical incentives range from $10,000 to $20,000 for energy efficiency and renewable energy measures.

Technical resources and design review. Focus on Energy has engineers and design professionals on staff who will participate in the design process, and will provide unbiased third-party feedback on design plans, construction documents, and specifications. Focus on Energy engineers will provide a list of recommendations on ways to achieve energy efficiency without adding a lot to the cost of the project.

Marketing resources. For qualifying projects, Focus on Energy can provide a press release and/or a case study that highlights the energy-efficient elements of a project. Developers can use this material for promoting their buildings.

Working with Developers

The multifamily new-construction program is currently consulting on ap-



This graph indicates that the earlier energy efficiency is incorporated into the design of a new project, the more cost-effective it becomes to implement.

proximately 100 projects throughout the state of Wisconsin. These projects range from small buildings of 8 to 12 units to very large, mixed-use projects with budgets well over $100 million.

Focus on Energy representatives can review architectural and Mechanical/Electrical/Plumbing (MEP) drawings and discuss the proposed project with the developer and other members of the design team. Working together with the project team member, a Focus on Energy representative will develop a list of potential energy efficiency measures customized for a particular project. From there, we collaborate with the design team to address any technical problems that might arise. After installation is verified by a Focus on Energy representative, the developer may qualify for cash incentives based on the type of measures that are installed.

Ideally, Focus on Energy becomes involved as early as possible in the design of the project. This makes it possible to achieve the largest potential savings for energy efficiency. In a typical new construction project, the first design decisions made, including site selection and building orientation, will dictate over 70% of the energy consumption of the building. As a result, the opportunities for energy and cost savings increase when Focus on Energy can be involved in the project as early as possible.

Once a project team has selected the types of systems that they are planning to use, we can make recommendations for increasing the efficiencies of the equipment. For example, when we work with a project that is still early in the design phase, we can discuss the benefits of certain heating systems, such as in-floor radiant heat. However, once the concrete for the floor has been poured, it generally is not feasible to consider this option. The last thing we, as an energy efficiency consulting group, want to hear is that we have great ideas but they've already purchased other, less efficient equipment. The consequences of thinking about energy efficiency too late in the game are lost opportunities and added costs. However, if Focus on Energy is involved early in the process, we can incorporate energy efficiency into the project at very little or no additional time or expense.

Promoting Energy Efficiency in Affordable Housing

After we published *Design & Build*, we contacted the Wisconsin Housing and Economic Development Authority (WHEDA) to promote the manual to multifamily professionals involved in affordable housing. It was a natural fit to address energy efficiency through WHEDA's low-income housing tax credit application process, because there is such a high demand for the tax credit program.

Under our collaboration with WHEDA, developers applying for low-income housing tax credits must contact Focus on Energy to learn how to incorporate cost-effective energy efficiency measures into their developments. Focus on Energy will provide WHEDA low-income housing tax credit applicants with a consultation, during which they can discuss their projects with a Focus on Energy representative. They will come away with a list of cost-effective energy efficiency and high-performance green building recommendations specific to their project. In addition, Focus on Energy will provide WHEDA applicants with copies of *Design & Build* at no charge. Projects that implement energy efficiency measures may also be eligible for cash incentives from Focus on Energy.

Documenting Success Stories

Design & Build includes a review of four multifamily projects that successfully implemented energy efficiency and renewable energy recommendations from Focus on Energy. These projects achieved improved operating costs and created more value for the occupant in reduced utility bills. They also created more value for owners in higher occupancy rates and increased value of the building. The case studies include information about the specific technologies that were employed, and they give contact information about the project teams.

Yahara River View Apartments is a 60-unit multifamily housing facility with 52 units dedicated to meeting the needs of low- and moderate-income families. Common Wealth Development, based in Madison, Wisconsin, remediated an industrial brownfield site and worked with neighbors to design a building that fits in with its community. The efficient

The *Design & Build* manual can help guide a multifamily development from the planning stages to resident move in.

design of this building extends to the floor plan, and to the use of durable, low-maintenance building materials. High-efficiency equipment and a natural gas submetering system that logs each unit's space-heating energy use and allocates appropriate costs to the residents contribute to low building operating costs and help to keep rents low.

The building envelope is built tight and is air sealed, with appropriate ventilation, using insulation and Energy Star-qualified windows and fans. A system of high-efficiency sealed-combustion condensing modulating boilers is networked together with two storage tanks. Solar panels on the roof provide heated water to another storage tank, which is connected to the boiler system. The solar domestic hot water system provides 20%–30% of the hot water used in the building each year, and up to 70 % on warm sunny days. The heating system is submetered and billed to residents based on actual usage. A typical heating bill for a one bedroom apartment (with an area of 667 to 783 square feet) was about $13 per month during the 2004–2005 heating season. Residents took notice—the apartments were 99% occupied last year—while the average vacancy rate in Madison was 7%–10%. Energy savings for the building are estimated at 72,200 kWh and 23,400 therms per year. The cost to build at Yahara came out at $63 per square foot.

Energy Efficiency and the Local Economy

According to the Wisconsin Department of Administration (*Legislative Update 2*, no. 4 [June 2005]: 00–00), the state of Wisconsin receives a net return of $6.56 for every $1.00 invested in Focus on Energy programs. Since Wisconsin must import virtually all of its energy from out of state, every dollar Wisconsin spends on energy efficiency is a dollar that supports the local economy and reduces a dollar that goes out of state to import electricity, coal, or natural gas. Keeping money in Wisconsin saves jobs, increases personal income, and makes our economy more efficient and competitive overall. The less Wisconsin businesses must rely on outside resources, such as imported coal or natural gas, the more resilient our local economy will be.

When multifamily experts collaborate on building healthy, energy-efficient buildings, everyone wins. Using *Design & Build* to plan, develop, and build multifamily buildings is an excellent way to help ensure that the state's economy remains strong, and that its residents are healthy and satisfied with their homes.

Josh Arnold, JD, MBA, LEED AP, manages the Multifamily New-Construction Program for Focus on Energy. He speaks nationally about sustainable development, high-performance green buildings, and energy efficiency. Energy Center of Wisconsin staff contributed to this article.

For more information:

Josh Arnold can be reached by telephone at (888)201-4061 or by e-mail at jarnold@franklinenergy.com.

To request a copy of *Design & Build for Energy Efficiency: Multifamily New Construction Best Practices Guide*, please visit the Focus on Energy Web site at www.focusonenergy.com.

The Advanced Building guidelines can be downloaded at no charge from www.poweryourdesign.com.

To learn more about the Energy Center of Wisconsin, go to www.ecw.org.

For more information on the New Buildings Institute, go to www.newbuildings.org.

For more information about the Wisconsin Housing and Economic Development Authority (WHEDA) please see their Web site at www.wheda.com.

The LEED for New Construction Version 2.2 Rating System (October 2005) can be viewed at no charge by going to www.usgbc.org/DisplayPage.aspx?CMSPageID=220#v2.2.